Supporting Information

NOSTOS Prevents Silent Acquisition- and Scale-Specific Failure in Quantitative Microscopy

Yan Jun Lin

This file reports estimator-level validations that support implementation breadth but are not pooled with the four main selective-validity experiments. Every value is read directly from a frozen JSON receipt; public images are not redistributed in the software archive.

S1. Synthetic truth and perturbation validation

Nine analytic constructs encoded programmed orientation, spectral scale, blob, tube, sheet, thickness, roughness, network structure and spatial heterogeneity. Eight primary perturbations were applied to the orientation-scale construct. The five module gates evaluated structure tensor, Hessian morphology, local thickness, network survival and directional spatial range. Mask dilation and erosion were evaluated as sensitivities because invariance to an altered mask is not scientifically expected.

Table S1. Exact programmed-perturbation results.

Perturbation	Magnitude	Angular error	Scale error	Gate
rotation	17	0.0001°	2.986%	pass
resampling	0.75	0.0001°	0.772%	pass
crop	0.8	0.0005°	0.211%	pass
blur	1	0.0000°	0.772%	pass
noise	0.1	0.0098°	0.772%	pass
contrast	0.65	0.0000°	0.772%	pass
psf	0.75	0.0000°	0.772%	pass
partial volume	0.7	0.0003°	0.772%	pass

Table S2. Frozen module-by-perturbation gate counts.

Module	Required	Passed	Mask sensitivities
tensor	5	5	—
hessian	12	12	—
geometry	2	2	2
network	1	1	—
spatial	4	4	—

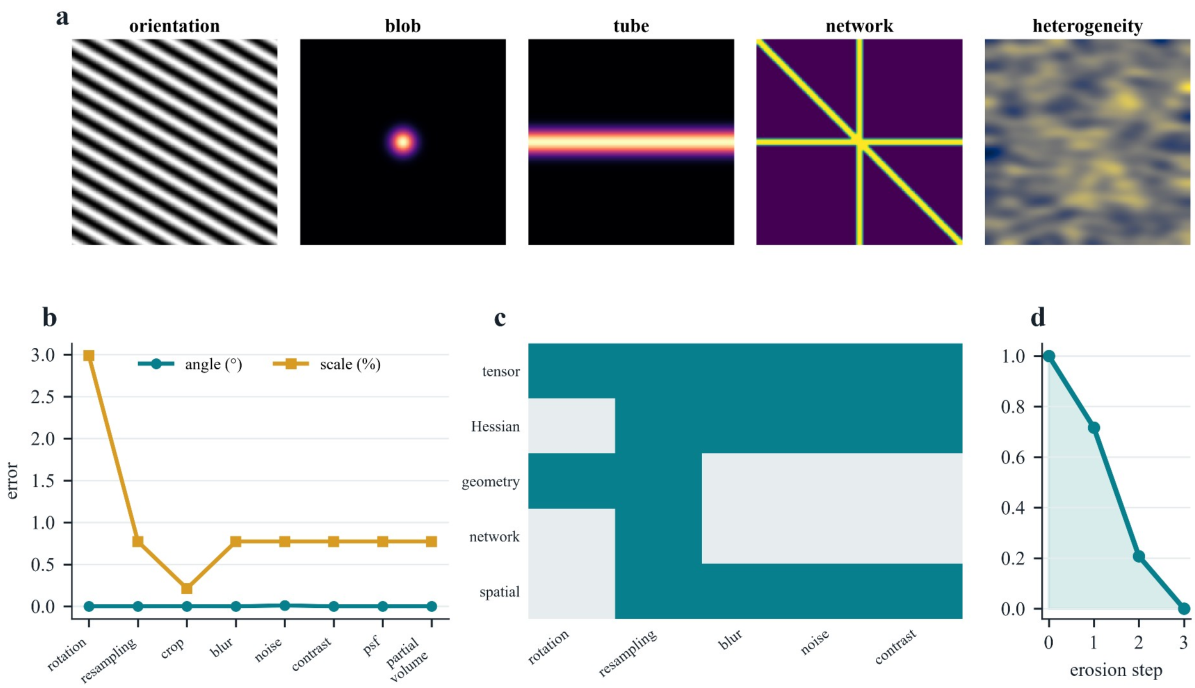

Figure S1. Synthetic operating-envelope validation. a, Programmed analytic constructs. b, Angular and physical-scale errors under the eight primary perturbations. c, Required module-by-perturbation cells; teal denotes a passed required cell and grey denotes a combination not declared by the protocol. d, Monotonic network survival under erosion. All 24 required module tests passed; the two mask-error experiments were retained as sensitivity analyses.

S2. External network measurement on STARE reference masks

Twenty STARE reference masks were analyzed as binary networks. NOSTOS and the independent scikit-image skeletonize comparator used the same mask support. Sampling changes preserved both network erosion survival and skeleton length. The result validates endpoint implementation on supplied masks; it does not validate automatic vessel segmentation or infer vascular biology.

Table S3. STARE network endpoint confirmation.

Endpoint	Independent images	Agreement	Additional diagnostic
erosion-survival AUC	20	Spearman $\rho = 0.988$	inter-annotator $\rho = 0.874$
skeleton length	20	Spearman $\rho = 0.995$	median relative error = 0.018

S3. External trabecular-bone local thickness

Eight public micro-CT volumes from Zenodo record 11061947 were analyzed in physical units. NOSTOS maximal-sphere local thickness was compared first with the archived IPL map and then with BoneJ 1.4.3 executed in a checksum-locked legacy Fiji environment. The independent image volume was the inferential unit. These comparisons establish agreement for local thickness on supplied bone masks, not diagnostic performance or tissue competence.

Table S4. Public trabecular-bone thickness comparisons.

Comparator	Volumes	Primary agreement	Error
archived IPL map	8	median voxelwise $\rho = 0.927$	mean MAE = 0.0189 mm
BoneJ 1.4.3	8	CCC = 0.926	mean $\lvert\Delta\rvert$ = 0.0157 mm

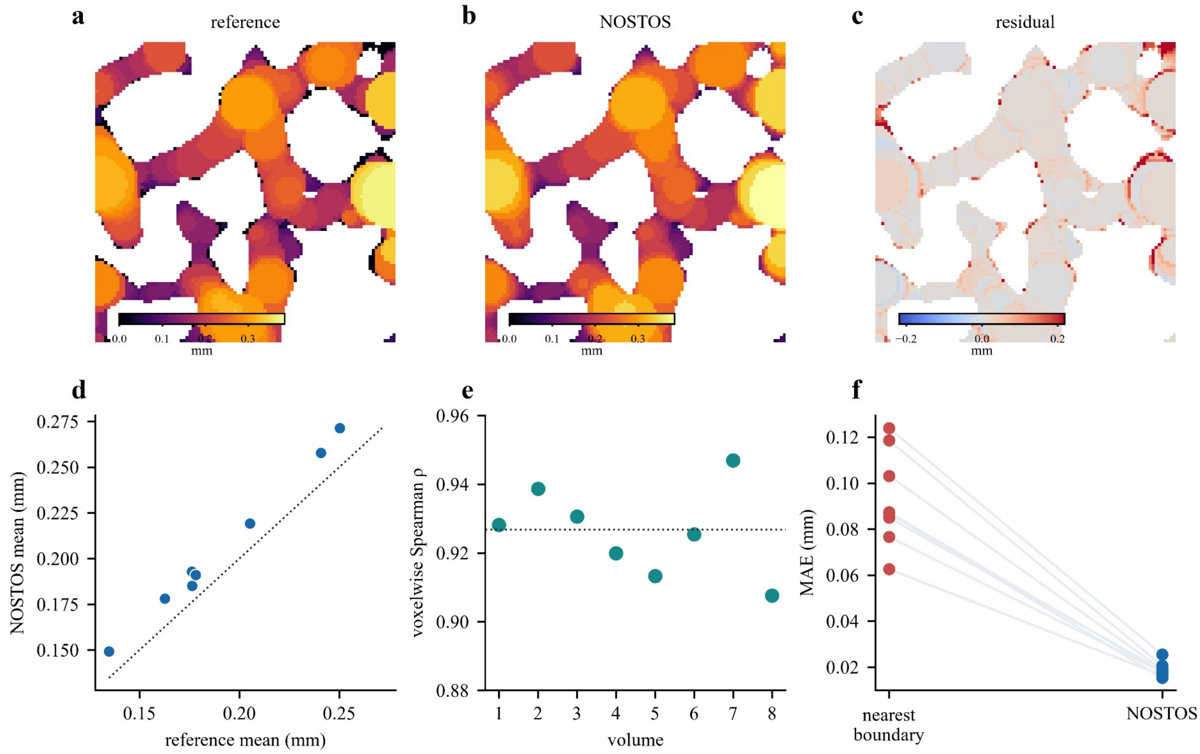

Figure S2. Public trabecular-bone local-thickness validation. a–c, Archived reference, NOSTOS estimate and residual on the same volume plane. d, Volume-level mean thickness. e, Voxelwise rank agreement in eight independent volumes. f, Paired mean absolute error for the nearest-boundary baseline and NOSTOS. The exact one-sided paired Wilcoxon value for the improvement over the baseline was 0.00390625.

S4. Dynamic deformation and spatial-response confirmations

Dynamic and spatial modules were tested separately so that their bounded operating claims could not be mistaken for biological validation. BBBC035 phase-correlation tests used programmed translations derived from public images. Dense deformation used programmed non-rigid fields. BBBC006 spatial responses compared adjacent focal planes with an intentionally defocused plane.

Table S5. Separate dynamic and spatial operating-envelope confirmations.

Module	Independent cases	Frozen result	Boundary
bulk registration	public-image series	maximum error = 0.0 px	programmed translation
dense deformation	8	median endpoint error = 0.270 px	comparator = 0.523 px
spatial response	64	adjacent-plane ρ = 1.000	defocus > adjacent in 100%

S5. Label-free and three-dimensional bone stress tests

The bone transfer program exercised orientation, network and scale support on mouse SHG, rat confocal lacuna-canalicular stacks, human synchrotron nanoCT and human ultraviolet photoacoustic microscopy. These analyses were intentionally kept as stress tests. A missing pixel calibration forced the UV-PAM spectrum to remain pixel-domain; insufficient support or unstable topology produced abstention rather than an inferred physical measurement.

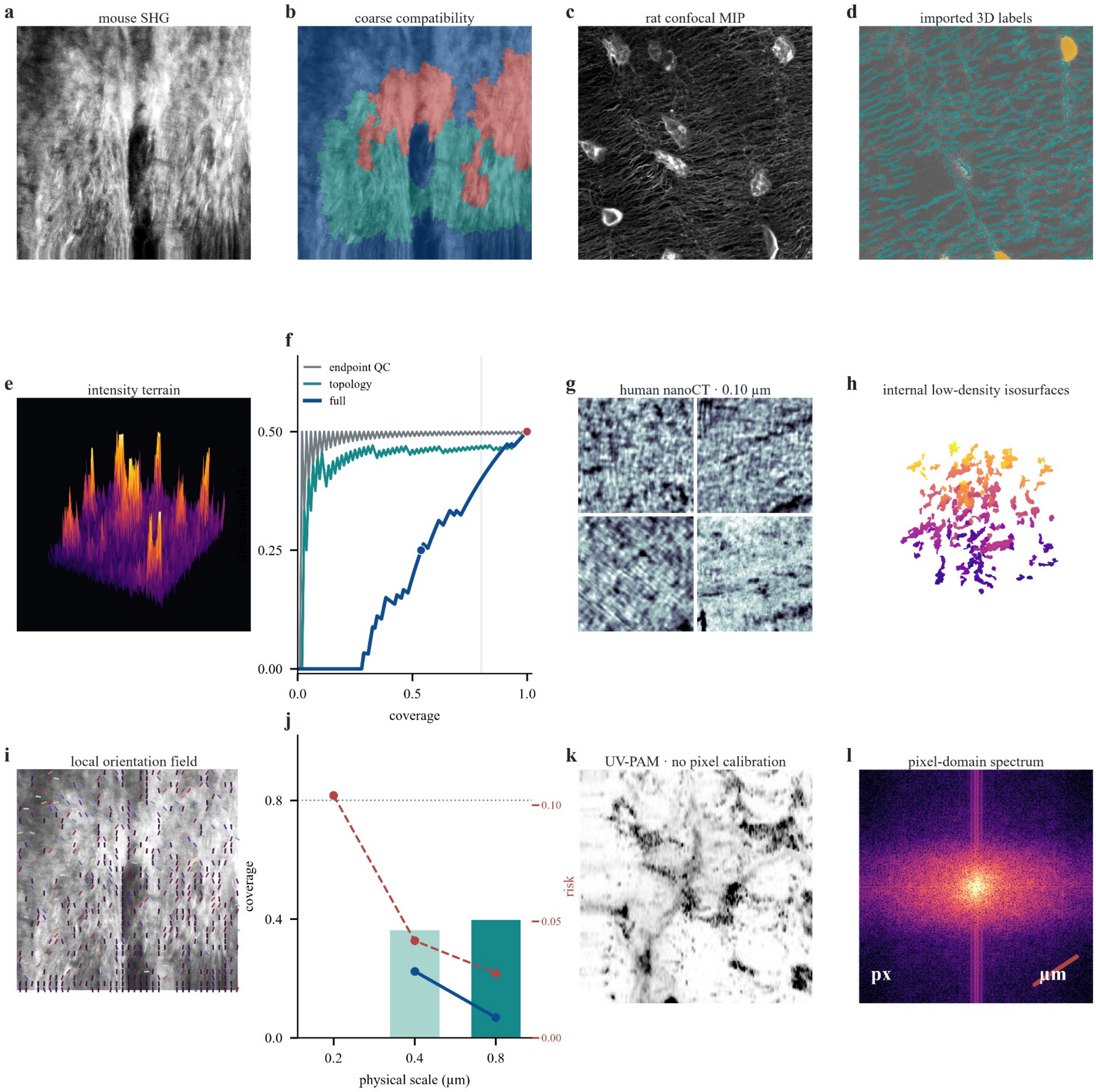

Figure S3. Public bone stress atlas. a,b, Mouse-bone SHG and coarse orientation-compatibility regions. c,d, Rat confocal projection and imported lacuna-canalicular labels. e, Deterministic intensity terrain. f, Risk–coverage curves for endpoint QC, topology-only support and the full contract. g,h, Human nanoCT crops and internal low-density isosurfaces. i,j, Local orientation and its scale-conditioned coverage/risk boundary. k,l, Human-bone UV-PAM and its pixel-domain Fourier spectrum. These exploratory transfers are not pooled with the primary confirmations.

S6. Frozen receipt index

The relative record name and SHA-256 digest below identify the exact machine-readable evidence used to generate this Supporting Information. The release builder verifies these records before packaging and excludes bulk public microscopy.

Table S6. Machine-readable evidence receipts.

Receipt	SHA-256
outputs/nostos0-synthetic-v1/validation.json	f406cce0826032ceb5d5146dc7e7181144ea7dd492ba2b950bd7abc9b98c146b
outputs/nostos0-module-perturbations-v1/module_perturbation_matrix.json	b8de6dacb398b20ba320f3163b4d5d43d5178ed5d5dda23e25c70f47463d44fe
outputs/nostos0-stare-network-confirmation-v1/ stare_network_confirmation.json	da9c59ad2508685325ab2ee805b94c23d90b5ce80a583cd51f5b41a0305fdb96
outputs/external-bone-v1/external_bone_validation.json	d0be3f11fb75dbd5221edf6659040e1cb043ddb710a32f66ab20533a05b4288f
outputs/nostos0-bonej-thickness-v1/bonej_thickness_comparator.json	561f5592e9d05f66240d1c17caa9f1fbd84d18a5af203cabaec37caf499a04f7
outputs/nostos0-bbbc035-dynamic-confirmation-v1/ bbbc035_dynamic_confirmation.json	373a1dfa7bcad32cf272be0b320e21ae5b9b4f9aa4d2c220f7118a5c862a94b4
outputs/nostos0-bbbc035-dense-deformation-confirmation-v1/ bbbc035_dense_deformation_confirmation.json	7babed3f53f28c6a2976169a94f5b17e8d182a5d3fd6f84538499d4e39c11f04
outputs/nostos0-bbbc006-spatial-confirmation-v1/ bbbc006_spatial_confirmation.json	511e1ed21391f082f221d64b76fb39b509b6e9d0f82e0c4848f4227fe91877fb

S7. Resource identifiers and scope

Table S7. External public resources.

Resource	Persistent identifier	Permitted interpretation
STARE	https://cecas.clemson.edu/~ahoover/stare/probing/	reference-mask network endpoints
trabecular bone	https://doi.org/10.5281/zenodo.11061947	local thickness on supplied masks
BBBC035	https://bbbc.broadinstitute.org/BBBC035	programmed bulk and dense motion
BBBC006	https://bbbc.broadinstitute.org/BBBC006	adjacent-plane and defocus response

No result in this file establishes diagnosis, mechanics, clinical usefulness, intraoperative performance, automatic tissue segmentation or a universal biological meaning for any NOSTOS coordinate. The supplementary results establish only the declared estimator-level agreements and perturbation behaviors.